



Testing and Training Analogical Responding in Young Children Using A Relational Evaluation Procedure

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Abstract

Analogical responding, or relating relations, as operationally defined by relational frame theory (RFT), is ubiquitous in everyday language and cognition, and is a key component in learning. However, to date, little research on the assessment and training of analogical relations in young children exists in the behavioral literature. The current research aimed to assess and train analogical responding in young, typically developing children. Three 5-year-old children were assessed and trained in relating relations using an RFT-based protocol in a multiple baseline design. Following multiple exemplar training, correct responding increased to criterion levels for all three children, and both generalization and maintenance were observed. Results are discussed in relation to previous research as well as the future testing and training potential of the RFT-based protocol for more complex language.

Keywords Analogy · Assessing analogy · Training analogy · Relational frame theory · Children

Analogy has been defined as “a comparison between two objects, or systems of objects, that highlights respects in which they are thought to be similar” (Bartha, 2019, Reasoning-Analogy section). Analogy is ubiquitous in daily language and cognition and is critical in various key domains of human activity including science, technology, and education (Bassok, 2001; Bod, 2009; Gentner, 1983; Hofstadter, 2001; Holyoak & Thagard, 1995; Morsanyi & Holyoak, 2010; Polya, 1945/2004; Sternberg, 1977; Stewart et al., 2004; Stewart et al., 2013). Furthermore, it is frequently used as a metric of intelligent behavior (Sternberg, 1977), and as a measure to predict academic success; for example, in the Law School Admissions Test (LSAT; Lapiana, 2004). As examples of everyday analogy consider phrases such as “she’s as busy as a bee” or “he’s like a bull in a China shop.” Examples of relatively well-known uses of analogy in educational or scientific contexts might include the comparison of electricity with water or of atomic structure with the solar system.

Considering the importance of analogy for intellectual development, cognitive-developmental psychologists have

examined the emergence of this skill in young children. Early researchers believed that analogical reasoning developed at the age of 12 or later, and that children younger than 12 years relied on simple associative strategies to solve analogies (Levinson & Carpenter, 1974; Lunzer, 1965; Piaget et al., 1977/2001; Sternberg & Nigro, 1980). In more recent research, however, it has been argued that dependent on a child’s previous experiences and knowledge, children as young as 3 or 4 can show analogical reasoning (Goswami & Brown, 1989, 1990; Richland et al., 2006). For example, Goswami and Brown (1989) examined analogical tasks based on relations of physical causality (e.g., ice: melting ice) and found that 3-year-old children could complete these tasks if they had relevant knowledge about the domains involved (see also Alexander et al., 1989; Goswami, 1989). However, in a subsequent replication of Goswami and Brown, Rattermann and Gentner (1998) concluded that children younger than 5 responded more frequently with matching responses instead of analogical responses.

Research on analogy has been mostly the province of cognitive psychologists; however, during the last two decades behavioral psychologists have also begun to research analogy. For example, relational frame theory (RFT) literature has explicitly recognized the theoretical importance of analogy (see e.g., Stewart et al., 2009). RFT is a contextual behavior analytic theoretical account that sees arbitrarily applicable derived relational responding (AADRR), or relational framing, as the

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key operant underlying human language and cognition (Hayes et al., 2001; Stewart, 2016; Stewart & Roche, 2013). According to this account, many nonhuman species can show nonarbitrarily applicable relational responding (NAARR), which involves relating stimuli based on their physical properties (e.g., selecting a stimulus physically similar, different, smaller, or larger than another; see, for example, Fagot & Maugard, 2013; Gillan et al., 1981; Smirnova et al., 2015). However, humans are unique in their additional ability to relate stimuli based on contextual cues that specify the relation rather than on the formal or physical properties of those stimuli. For example, if a child learns that X is greater than Y, they can then derive the untaught relation Y is less than X without seeing any physical objects. RFT sees this arbitrarily applicable derived relational responding as a generalized operant behaviour acquired via multiple exemplar training (MET) provided through typical exposure to a child's socioverbal environment. Coordinate (sameness) framing or equivalence, as originally demonstrated by Sidman (1971), is the earliest and most researched example of AADRR. Subsequent research has provided evidence of various other patterns of framing including, for example, difference (e.g., Steele & Hayes, 1991), opposition (Barnes-Holmes et al., 2004a), comparison (Dymond & Barnes, 1995), hierarchy (Gil et al., 2014), analogy (Barnes et al., 1997), temporality (O'Hora et al., 2005), and deixis (McHugh et al., 2004), among others.

From an RFT perspective, relational framing is defined by the three properties of mutual entailment, combinatorial entailment, and transformation of stimulus function, which together facilitate the generativity and flexibility of human language and cognition (Hayes et al., 2001). Mutual entailment is the property of bidirectionality of stimulus relations whereby if A is related to B, then B will be related to A (e.g., if A is opposite to B then B is opposite to A). Combinatorial entailment involves the combination of previously acquired mutually entailed relations to allow derivation of novel relations (e.g., if A is opposite to B, and B is opposite to C, then A is the same as C, and C is the same as A). Transformation of stimulus functions is the property whereby the psychological functions of a stimulus in a derived relation change depending on the nature of the relation and the functions of related stimuli. For example, if a child has already learned that A has monetary value and then derives that a novel coin C is worth more than A, they will choose C over A despite the fact that C is novel. On the other hand, if an arbitrary stimulus A has acquired an aversive function (e.g., by being paired with shock), then despite not being directly paired with shock itself, C might become more aversive than A based on the derived "larger than" relation. Indeed, previous research has shown such an effect (e.g., Dougher et al., 2007).

Empirical analyses of relational framing as a generalized operant show that it can be trained with targeted multiple exemplar training if it is weak or absent (Barnes-Holmes

et al., 2004a; Barnes-Holmes et al., 2004b; Berens & Hayes, 2007; Dymond et al., 2010; McHugh et al., 2004; Steele & Hayes, 1991; Stewart, 2016). For example, Barnes-Holmes et al., (2004b) successfully trained comparative AADRR in 4- to 6-year-old children. The researchers used two or three identically sized paper coins of assorted colors to test and train patterns of relational responding in accordance with more-than and less-than relations. During the experiment, they explained to the children how the coins compared to each other in terms of their value, and then asked them to pick the coin that would buy the most candy. Using MET, they successfully established more-less relations as generalized operant behavior by reinforcing correct answers (see also, Berens & Hayes, 2007; Gorham et al., 2009).

More recent studies have shown that training relational framing also affects general cognitive ability. For example, Cassidy et al. (2011) investigated the effects of an automated multiple-exemplar relational training program for frames of coordination, opposition, and comparison. Participants included typically developing 10- to 12-year-old children, several of whom were experiencing educational difficulties in school. Cassidy et al. found significant increases in IQ scores for all participants who received relational training. These data clearly indicate that cognitive ability is strongly related to one's repertoire of derived relational responding. Several replications of Cassidy et al. (2011) have shown similar increases in IQ scores following relational training (e.g. Cassidy et al., 2016; Colbert et al., 2018; Hayes & Stewart, 2016).

The studies just reviewed constitute part of a growing body of evidence suggesting that relational framing is the principal behavior characterizing human language and cognition, such that to understand how humans acquire relational framing is to understand how they acquire language (Dymond & Roche, 2013; Fryling et al., 2020; Zettle et al., 2016). One other source of evidence is research that has used laboratory-controlled patterns of relational framing to model and investigate analogy as one particular subtype of language. Working within an RFT framework, Barnes et al. (1997) provided the first functional analytic definition of analogy as the derivation of a sameness or equivalence relation between derived relations. For instance, consider the analogy *peach* is to *pineapple* as *goat* is to *horse*. In this case, *peach* and *pineapple* participate in an equivalence relation in the context of fruit, and *goat* and *horse* participate in an equivalence relation in the context of animal; thus, because these are both equivalence relations, we can derive a relation of equivalence between the relations themselves.

To model this phenomenon, Barnes et al. (1997) first trained and tested 4 three-member equivalence relations in adults and 9-year-old children. A matching-to-sample (MTS) procedure was used to train conditional discriminations among three-letter nonsense syllables (coded using alphanumeric designations) as follows: A1→B1, A1→C1, A2→B2,

A2→C2, A3→B3, A3→C3, A4→B4, A4→C4. Next, participants were tested for the derivation of the following four untrained combinatorially entailed relations: B1↔C1, B2↔C2, B3↔C3, B4↔C4. After participants passed equivalence tests, they were tested for the derivation of equivalence relations between equivalence relations themselves, which the researchers referred to as equivalence–equivalence responding. Compound stimuli comprised either two nonsense syllables that were equivalent or two that were nonequivalent. Participants were required to choose an equivalent pair in the presence of an equivalent pair (i.e., equivalence–equivalence) and a nonequivalent pair in the presence of a nonequivalent pair (i.e., nonequivalence–nonequivalence). For example, given B3C3 and B3C4 as comparisons, if the sample was B1C1 then they had to choose B3C3, whereas if B1C2 was the sample, they had to choose B3C4. All participants (i.e., both the adults and the 9-year-old participants) successfully related relations, and thus this constituted a basic model of analogy.

Several studies since Barnes et al. (1997) have extended this model of analogy in a number of respects (Barnes-Holmes et al., 2005; Carpentier et al., 2002, 2003, 2004; Ruiz & Luciano, 2015; Stewart et al., 2001). One strand of research of particular interest for the present purpose focused on analogical responding in young children (e.g., Carpentier et al., 2002, 2003). Carpentier et al. (2002) found that adults and 9-year-old participants readily showed equivalence–equivalence (i.e., as in Barnes et al., 1997), but 5-year-old children, although readily passing equivalence, initially failed to show equivalence–equivalence without additional prompting. In particular, the 5-year-old children required pretesting with compound–compound matching tasks involving trained (as opposed to derived) relations (e.g., A1B1–A3B3 and A1B2–A1B3) before they successfully passed the derived compound relations (BC–BC) test. Carpentier et al. (2003) extended this work by examining if additional compound–compound testing would also facilitate equivalence–equivalence performance in the absence of prior equivalence tests as had been seen in older participants; however, only 2 of the 18 five-year-old participants successfully passed this task.

The present study sought to extend Carpentier et al. (2002) and further examine the acquisition of analogical responding operationalized as derived relations between relations in 5-year-old children. It extended the earlier work first by attempting to directly train the skill of deriving relations between relations using a controlled multiple baseline design to demonstrate experimental control. In addition, in the present study, we used an alternative, more efficient format to assess and train the derived relational pattern than the match-to-sample (MTS) procedure used by Carpentier et al. Although often used in studies of derived relations, MTS procedures require extensive baseline training before any testing or training can begin. For example, in Experiment 1 of Carpentier

et al. (2002), the 5-year-old participants required an average of 234 baseline trials before training and testing could begin.

The RFT-based relational evaluation procedure (REP; see Barnes-Holmes et al., 2001; Stewart et al., 2004) offers a potential alternative to MTS. In the REP participants are required to evaluate relational networks based on the presentation of contextual cues juxtaposed with relevant stimuli. Using this protocol multiple exemplars of relational tasks can be readily generated and presented without requiring lengthy prerequisite training. The REP has been successfully employed in several recent RFT-based studies to train relational framing in young children. For example, in the previously mentioned Cassidy et al. (2011) study, an automated AADRR assessment and training program was based on the REP format (see also Cassidy et al., 2016; Hayes & Stewart, 2016). The automated program presented multiple exemplars of relational statements involving nonsense words juxtaposed with contextual cues (e.g., “CUG is the SAME as DAX,” “DAX is the SAME as YIM”), followed by questions requiring relational derivation based on those statements (e.g., “Is DAX the SAME as CUG?,” “Is CUG the same as YIM?”). The successful use of this protocol by Cassidy et al. to efficiently train key patterns of relational framing to 8- to 12-year-old children, which boosted their intellectual performance, suggests the potential utility of the REP format in training relational framing in children and hence we adopted this protocol in the current study.

One recent study in our lab has already successfully developed and tested a type of REP specially created to allow for assessing relational framing in young children. Kirsten and Stewart (2021) designed a comprehensive relational assessment to test a variety of relational frames across four levels of responding including nonarbitrary relations, nonarbitrary analogical relations, arbitrary relations, and arbitrary analogical relations. Assessment of the latter two stages of responding (i.e., arbitrary relations and analogical relations) was also required in the present study. To test these stages, Kirsten and Stewart taught children to respond to relational networks composed of monochromatic circles (instead of nonsense words) juxtaposed with single letters as the relevant contextual cues (e.g., S for *sameness*, D for *difference*). For example, children were taught that given a green circle and a red circle separated by the contextual cue “S” they should subsequently treat the green and red circles as equivalent. In the analogical relations stage, the analogical stimuli were compound elements composed of monochromatic circles presented below the relational network. This format allowed young children not yet able to read to report on and evaluate multiple exemplars of arbitrarily applicable relational networks defined by specifically selected contextual cues. This methodology was effective in assessing arbitrary relational responding as well as the relating of arbitrary relations in these children.

The aim of the present study was to extend the work of Carpentier et al. (2002) as well as that of Kirsten and Stewart

(2021) by not just assessing but also training the relating of relations in a multiple baseline design with 5-year-old children using an adaptation of the REP type protocol used in Kirsten and Stewart.

Method

Participants and Setting

Participants were three typically developing children enrolled in a private elementary school on the east coast of the United States. P1 was a female aged 5 years, 11 months, P2 was a male aged 5 years, 7 months, and P3 was a male aged 5 years and 9 months. Participants were selected for inclusion based on their performance on an adaptation of the relational assessment used in Kirsten and Stewart (2021; see Fig. 1 for a summary version of the preassessment). All probe and training sessions were administered by the researcher in an otherwise unoccupied classroom of the participants' school during school hours. The researcher sat next to the participant at a standard school desk. A second, independent observer sat approximately 3 feet away from the desk on the other side of the participant with a full view of the participant and the computer screen.

Ethical approval for recruitment of participants was obtained from the research ethics committee of the lead researcher's

host institution. Consent for conducting the study was also obtained from the principal of the school. Caregiver consent was obtained for each child who participated, and verbal assent was obtained from each of the participants.

Experimental Design

A combination multiple baseline design across participants and a multiple probe design across behaviors was used in this study. The study included a relational preassessment for screening potential participants; a baseline condition in which analogy was tested; and a training condition in which analogical responding was trained and generalization probe trials were presented. In order for P2 to enter the training condition, the participant previously exposed to training (i.e., P1) had to meet the training criterion (i.e., scoring 100% correct twice consecutively on the trained repertoire) whereas for P3 to enter the training condition, P2 had to meet the training criterion. In this way, the design controlled for coincidental extraneous events such as baseline assessment, maturation, and so on (see e.g., Hayes, 1981).

Materials and Apparatus

A 13" MacBook running Microsoft PowerPoint was used to present trials. Individual stimuli included colored circles (approximately 1 in. in diameter) and the letters "S" for *sameness*

RELATIONAL SECTION

#	Slide 1: Same-Different	R
1	S means the same	1
2	What does S mean?	same
3	D means different	2
4	What does D mean?	diff
5	Point to same	
6	Point to different	
7	What does this (S) mean?	same
8	What does this (D) mean?	diff

Contextual cues: *sameness* & *difference*

#	Slide 2: Red is the same as blue	R
1	Is red the same as blue?	yes
2	Is blue the same as red?	yes
3	What is red to blue?	same
4	What is blue to red?	same
5	Are red and blue the same or different?	same
6	Is red different to blue?	no
7	Is blue different to red?	no
8	What is the same as red?	blue
9	What is the same as blue?	red
10	What does S mean?	same

Relational network: red is the same as/different to blue. Directly presented (DP) and mutual entailment trials (ME)

#	Slide 3: Red is different to blue	R
1	Is red different to blue?	yes
2	Is blue different to red?	yes
3	What is blue to red?	diff
4	Are red and blue the same or different?	diff
5	Is red the same as blue?	no
6	What is different to blue?	red

DP, ME, and combinatorial entailment (CE) trials

#	Slide 4: Red S Blue, Blue S Yellow	R
1	Is red the same as blue?	yes
2	Is blue the same as yellow?	yes
3	Is red the same or different to yellow?	same
4	What is yellow to red?	same
5	Is yellow the same as red?	yes
6	Is yellow different to red?	no
7	What is red to yellow?	same
8	What is the same as yellow? (2 R)	red/blue

Three Prompt Levels

#	Slide 5: Red S Blue, Blue D Yellow	R
1	Is red the same or different to blue?	same
2	Is blue the same or different to yellow?	diff
3	What is red to yellow?	diff
4	What is yellow to red?	diff
5	What is different to yellow? (2 R)	red/blue

ANALOGY SECTION

#	Slide 6: CE Equivalence Prompts S S S	R
1	Is blue the same or different to red?	same
2	Is yellow the same or different to blue?	same
3	Is green the same or different to yellow?	same
4	What is red to yellow?	same
5	Is green the same or different to blue?	same
6	What is yellow to red, same or different?	same
7	What is red to yellow?	same
8	What is red to green?	same
9	What is green to red?	same

Compound Trials

#	Slide 7: CE Equivalence Prompts S S D	R
1	Is blue the same or different to red?	same
2	Is yellow the same or different to blue?	same
3	Is green the same or different to yellow?	diff
4	Is blue the same or different to green?	diff
5	What is green to blue?	diff
6	What is red to green, same or different?	diff
7	What is green to red?	diff

#	Slide 9: Red S Blue, Blue D Yellow	R
1	What is blue to yellow, same or different?	diff

#	Slide 10: Red S Blue, Blue D Yellow	R
1	What is red to yellow, same or different?	diff

#	Slide 14: CE Analogy w/ Equiv Prompt (Viz + Voc)	R
1	Is red the same or different to green?	diff
2	Is blue the same or different to green?	diff
3	Is red the same or different to yellow?	same
4	Left: blue to green	

Relational network presented first in Level 1 & 2

slide	CE Analogy w/ Equiv Prompt (Viz)	R
26	Right: purple to green	
28	Right: blue to yellow	
30	Left: red to green	
32	Left: orange to purple	
34	Right: orange to yellow	

Analogical task

slide	CE Analogy No Prompt	R
36	Right: orange to green	
37	Left: blue to yellow	
38	Left: green to red	
39	Right: blue to yellow	
40	Left: orange to purple	

Fig. 1 Preassessment: Sample of Relational and Analogical Sections. *Note.* In the CE Analogy w/ Equiv. Prompt (Viz & Voc) trials, the relational network was first presented without the analogical stimuli plus three CE relational trials followed by analogical stimuli; in the CE Analogy w/ Equiv. Prompt (Viz) trials, the relational network was first

presented without the analogical stimuli followed by the analogical stimuli; in the CE Analogy No Prompt trials, the relational network and the analogical stimuli were presented on the screen simultaneously. Adapted from Kirsten and Stewart (2021)

and “D” for *difference* relations in Calibri or Arial, size 40 font (see first panel in Fig. 2). During the relational section of the preassessment, trials included an array of such stimuli at the center of the screen that were designated by the experimenter as participating in a relational network. During the analogical section of the preassessment, as well as during the study probe and training sessions, similar to the relational section, trials included an array of stimuli that were designated as participating in a relational network. In this section, however, these appeared in the top portion of the screen whereas in the bottom portion of the screen there appeared a sample compound element below the relational network; and two comparison compound elements below the sample on the bottom left and right of the screen, separated by a black line (see third panel in Fig. 2 for an illustrative example of the relational network and compound elements). The relational network in the top portion of the screen included six monochromatic circles and the relational cues, S for *same* and D for *different*, to delineate the relations between the circles (see second panel in Fig. 2). For example, one possible array might be represented as follows: [Red Circle] [S] [Blue Circle], [Blue Circle] [S] [Yellow

Circle], [Yellow Circle] [D] [Green Circle], which indicates that the red circle is the same as the blue circle, the blue circle is the same as the yellow circle, and the yellow circle is different from the green circle. The compound elements that appeared lower down the screen in black, outlined rectangles as the sample and comparison stimuli were each composed of two of the monochromatic circles from the relational network but did not contain relational cues (see third panel in Fig. 2). For example, one such compound might be designated as [Red Circle][Yellow Circle]. Each slide had only one relational network and one set of analogical task stimuli.

Procedure

Overview

The following will provide procedural details of the relational preassessment and the multiple baseline component of the study. The preassessment was administered first in one session and took 5–25 min to complete. Following the preassessment, participants entered the multiple baseline

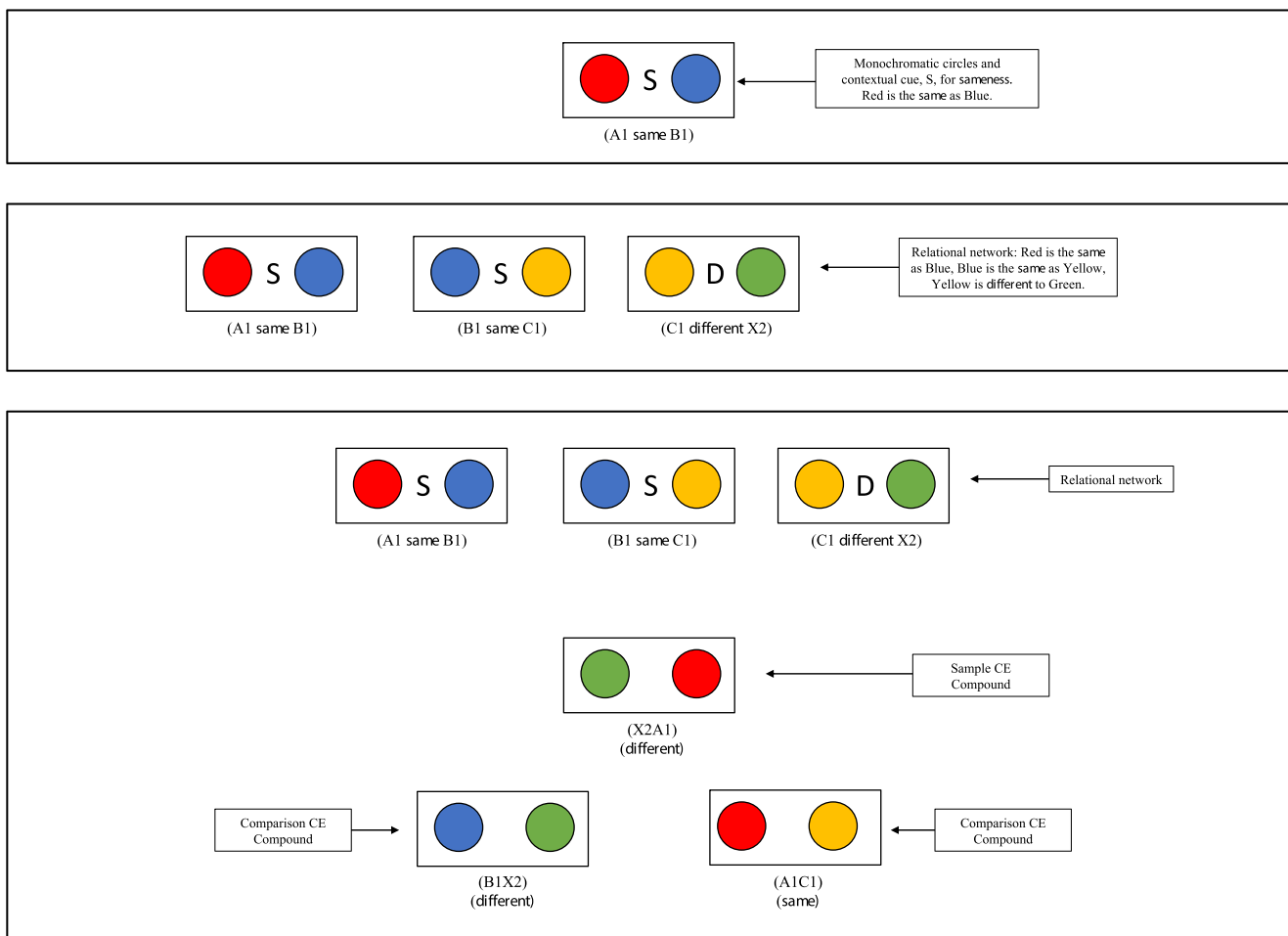


Fig. 2 Stimuli Format in Preassessment and Multiple Baseline Conditions. *Note.* First panel: monochromatic circles plus contextual cue, S; second panel: relational network; third panel: relational network plus analogical stimuli

including baseline sessions followed by training and multiple probe sessions. Sessions lasted 10–40 min, and on average included seven probe and training sets; a probe set took approximately 2–4 min to complete, and a training set took approximately 1–3 min to complete. All participants started baseline sessions on the same day, and all participants took approximately 3 weeks to complete the study. Maintenance probe sessions were administered 1 month after each participant's last probe in the training condition was administered.

Preassessment

The relational preassessment adapted tasks from the arbitrary relational and analogical stages in Kirsten and Stewart (2021). The preassessment included two main sections; the first section tested for derived relational responding including mutual entailment and combinatorial entailment across five subsections increasing in difficulty. The second section tested for derived relations between combinatorially entailed relations (i.e., analogical responding) across three levels of prompting (refer to Fig. 1 for a sample of trials and stimuli found in the preassessment). There were a total of 89 trials in the preassessment.

Section 1: Relational Tasks The first six trials in the preassessment introduced the participant to the contextual cues themselves (i.e., S and D; refer to Fig. 1). Next, the researcher introduced the participant to simple, arbitrary relational networks. The participant was shown a computer screen displaying a relational network, for example: [Red Circle] [S] [Blue Circle]. The assessor instructed the participant to look at the screen and said, “Let’s *read* this: Red is the same as Blue” (in this example and hereafter, *reading* refers to vocally identifying the stimuli and relational cues in the relational network in sequence from left to right, similar to textual reading). After delivering the instruction, the assessor asked 16 yes/no and same/different questions about the relational networks to which the child was required to respond vocally, including questions about directly presented relations (e.g., “Is Red the same as Blue?” or “Is Red the same or different to Blue?”), and questions requiring reversal of the directly presented relation (which we will hereafter refer to as mutual entailment; e.g., “Is Blue the same as Red?”). Participants were required to respond within approximately 5 s of the verbal instruction in these 22 trials; failure to do so was considered an incorrect response.

The next set of 29 trials included more than two stimuli, and questions became increasingly difficult and required responding not only to “directly presented” (DP) and “mutual entailment” (ME) questions but also to questions that required the combination of directly presented relations (hereafter referred to as combinatorial entailment; CE). The first set of questions in this section referred to a relational network in

which only two *sameness* relations were presented; the second set referred to a relational network including both a *sameness* and *difference* cue; the third set included three *sameness* relations, and the fourth set included the relational network format that would be used in the study training sets involving the sequence: A is the *same* as B, B is the *same* as C, and C is *different* to D. An example of the latter set might be as follows: The relational network [Red Circle] [S] [Blue Circle], [Blue Circle] [D] [Yellow Circle] is presented on the screen and then questions regarding combinatorially entailed relations (e.g., “Is Red the same/different to Yellow?”) are presented to which the child had to respond by vocally producing the correct relational term. Participants were required to respond within approximately 30 s of verbal instruction for this set of trials; failure to do so was considered an incorrect response.

The next set of questions involved compound elements as described in the previous section, a stimulus composed of two, side-by-side monochromatic circles from the relational network without the relational cues, S or D. For example, one such compound might be designated as [Red Circle][Yellow Circle]. In each trial, a relational network was presented at the top of the screen. Below the network, a white rectangle with a black outline contained the compound element (i.e., two differently colored circles identical to two of the circles in the relational network). The researcher and the participants read the relational network together and then the researcher said, “Look here [points to compound element], are they the same or different? Remember to look here [points to relational network] to help you figure it out.” Participants had to refer to the network in order to correctly identify the relation between the stimuli [the circles] in the compound element. For example, the relational network [Red Circle] [S] [Blue Circle], [Blue Circle] [D] [Yellow Circle] is presented at the top of the screen, and the compound stimulus (e.g., [Red Circle][Yellow Circle]) is presented below the network. Presented with this network, the participant might derive that red is different from yellow. Each slide had only one relational network and one set of analogical stimuli. Once again, the child had to respond by vocally producing the correct relational term. Participants were required to respond within approximately 30 s following the verbal instruction for these 22 trials; failure to do so was considered an incorrect response.

In order to proceed to the analogical questions in the preassessment, the participant had to pass the last 14 combinatorial entailment and compound identification trials in the relational section of the preassessment at 80% correct. Only the last 14 trials of the relational section were included because the previous sections introduced the participant to the format and did not exclusively ask combinatorially entailed or compound questions.

Section 2: Analogical Tasks In the analogical section of the preassessment, the analogical trial stimuli (i.e., both the

sample and two comparisons) included a relational network and three compound elements composed of two circles (see third panel in Fig. 2). For example, a presented relational network might be: [Red Circle] [S] [Blue Circle]; [Blue Circle] [S] [Yellow Circle]; [Yellow Circle] [D] [Green Circle]; and read as, “Red is the same as blue, blue is the same as yellow, and yellow is different from green.” The sample compound element presented below the relational network might be [Red : Green] and the comparison compound elements below the sample might be [Red : Yellow] and [Blue : Green]. It should be noted that in all trials both of the comparison compounds included one of the elements from the sample compound so that it was not possible to respond correctly based purely on nonarbitrary similarity. On each trial, the researcher read the relational network to/with the participant and then delivered the instruction, “Look at this one at the top [pointing to the sample compound]. Which one of these [pointing to each of the comparison compounds in turn] is like this one at the top?” Given the above relational network of [Red Circle] [S] [Blue Circle]; [Blue Circle] [S] [Yellow Circle]; [Yellow Circle] [D] [Green Circle], participants could derive the relation of *difference* for the sample compound [Red : Green], a relation of *sameness* for comparison stimulus [Red : Yellow], and a relation of *difference* for comparison stimulus [Blue : Green]. The correct (analogical) choice would be to select the second comparison (i.e., [Blue : Green]) because, as in the case of the sample stimulus, this involves a *difference* relation. Correct responses included either pointing to the correct comparison compound element or vocally stating the two colors of the correct comparison compound element.

For all trials in the preassessment and in the multiple baseline conditions, the assessor first read the relational network to/with the participant and then asked the trial questions. If the participant requested or initiated independent reading when presented with the relational network, the researcher did not read with the participant.

The analogical section of the preassessment included three levels of prompting, which together represented a “most-to-least” type prompting strategy. Each level included five combinatorially entailed (CE) analogy trials. Level 1 included CE analogies with visual and vocal equivalence prompts; Level 2 included CE analogies with visual prompts; and Level 3 included CE analogies with no prompts. Each level included five analogical trials. The procedure described in the previous paragraph above was used in Level 3 (CE analogy with no prompts). In Levels 1 and 2, the relational network was first presented on the screen without the compound elements (i.e., the sample and the comparisons) in order to ensure that the participant attended to the relational network. The researcher and participant read the relational network together. In Level 1, after reading the relational network, the researcher asked three combinatorial entailment questions that would refer to

the relations of the compound elements in the ensuing analogy task. That is, the CE questions were included as a vocal prompt to operate on the final, analogical response which required deriving relations within relations. For example, given the relational network [Red Circle] [S] [Blue Circle]; [Blue Circle] [S] [Yellow Circle]; [Yellow Circle] [D] [Green Circle], the researcher presented the three CE questions relevant to the analogical stimuli (not yet presented on the screen but known to the researcher). For example, if the analogical stimuli included the sample [Red : Green], and comparisons [Red : Yellow], and [Blue : Green], the researcher asked the CE questions: Is Red the same or different to Green? Is Red the same or different to Yellow? Is Blue the same or different to Green? After responding to the three CE questions, the analogical stimuli were presented below the relational network. The participant was given the instruction to look at the sample first, and then to choose which comparison was like the sample. Thus, in Level 1, each trial included three CE questions in addition to the analogy trial. Participants were required to respond within approximately 30 s following the CE questions, and within 1 min for the analogical tasks.¹ Failure to do so in either case was considered an incorrect response. Feedback was provided for the CE questions; correct responses were praised, and incorrect responses were re-presented until the participant emitted a correct response. Feedback was not provided for the analogical questions. In Level 2, the researcher and participant read the relational network together first (there were no combinatorial entailment questions) and then the analogical compound stimuli were presented below the relational network as in Level 1. In Level 3, as described earlier, the relational network and the analogical stimuli were presented simultaneously, and the researcher and participant read the relational network together first (there were no combinatorial entailment questions). Participants were required to respond within 1 min for all analogical trials; failure to do so was considered an incorrect response. All levels were presented regardless of performance in order to confirm that the potential participant could derive combinatorially entailed relations but could not complete analogical tasks (i.e., relate relations).

The passing criterion was 100% (5/5) correct analogy trials at each level. Hence, the passing criterion for the full analogy section of the preassessment was 15/15 analogy trials. With regard to inclusion as a whole, potential participants were required to pass the last set of relational trials in the preassessment (14 trials) at 80% or more, and to fail the analogy section (see Table 1 for a schematic presentation of the procedure).

¹ It was recognized in pilot work with adults that participants required time to complete these analogical tasks.

Multiple Baseline Conditions

The multiple baseline components included a baseline condition including unreinforced probe sessions, and a training/testing condition in which training and probe sessions were administered. The study included three different types of probes including Combinatorially Entailed Analogy Probes (CE Probes); Directly Presented/Mutually Entailed/Combinatorially Entailed Analogy Probes (DMC Probes); and *Difference-Cue-Shifted* Analogy Probes (D-cue Probes; see Fig. 3 for an illustrative example of each probe type).

In CE Probe sets, the relational network always included the presented relations A is the *same* as B, B is the *same* as C, and C is *different* to D. Combinatorial entailment was involved in deriving the relation between the two stimuli (i.e., two monochromatic circles) in all sample and comparison compound elements.

During baseline sessions, the same CE Probe set was administered for all sessions. Novel CE Probe sets were included for generalization testing (see first panel in Fig. 3).

The relational network in the DMC Probes followed the same pattern as in the CE Probe (i.e., [A [S] B], [B [S] C], [C [D] D])). However, the relations between the stimuli in the sample and comparison compound elements included a mixture (within as well as across trials) of directly related, mutually entailed, and/or combinatorially entailed relations. In directly related compounds, the relation between the stimuli was directly presented in the network and thus did not require any derivation (e.g., [AB] or [BC]). For example, a DMC Probe trial might include a sample with directly related stimuli, one comparison involving a mutually entailed relation, and a second comparison involving a combinatorially entailed relation (see second panel in Fig. 3).

Table 1 Schematic Presentation of the Preassessment and Multiple Baseline Conditions

Pre-Assessment		
Phase	Relational Targets	Passing Criteria %
Inclusion Criteria	1. Identifying contextual cues	100
	2. Reading relational networks	100
	3. Testing DP and ME relations	100
	4. Testing CE relations	80
	5. Testing compound elements	80
	Analogy Probes	
	6. Level 1: 3 CE trials + 5 analogy trials	100
	7. Level 2: 5 analogy trials	100
	8. Level 3: 5 analogy trials	100
Multiple Baseline Study		
Training Condition & Probes	Baseline: CE Probe Set 1	100
	• Training: 3 CE trials + 1 analogy trial	100
	• CE Probe Set 1: 5 trials	100
	• Novel CE Probe: 5 trials	100
	• DMC Probe: 5 trials	100
	• Novel CE Probe: 5 trials	100
	• D-Cue Probe: 5 trials	100
One-Month Maintenance Probes	Novel CE Probe	100
	Novel DMC Probe	100
	Novel D-Cue Probe	100

Note. DP = directly presented relations; ME = mutually entailed relations; CE = combinatorially entailed relations.

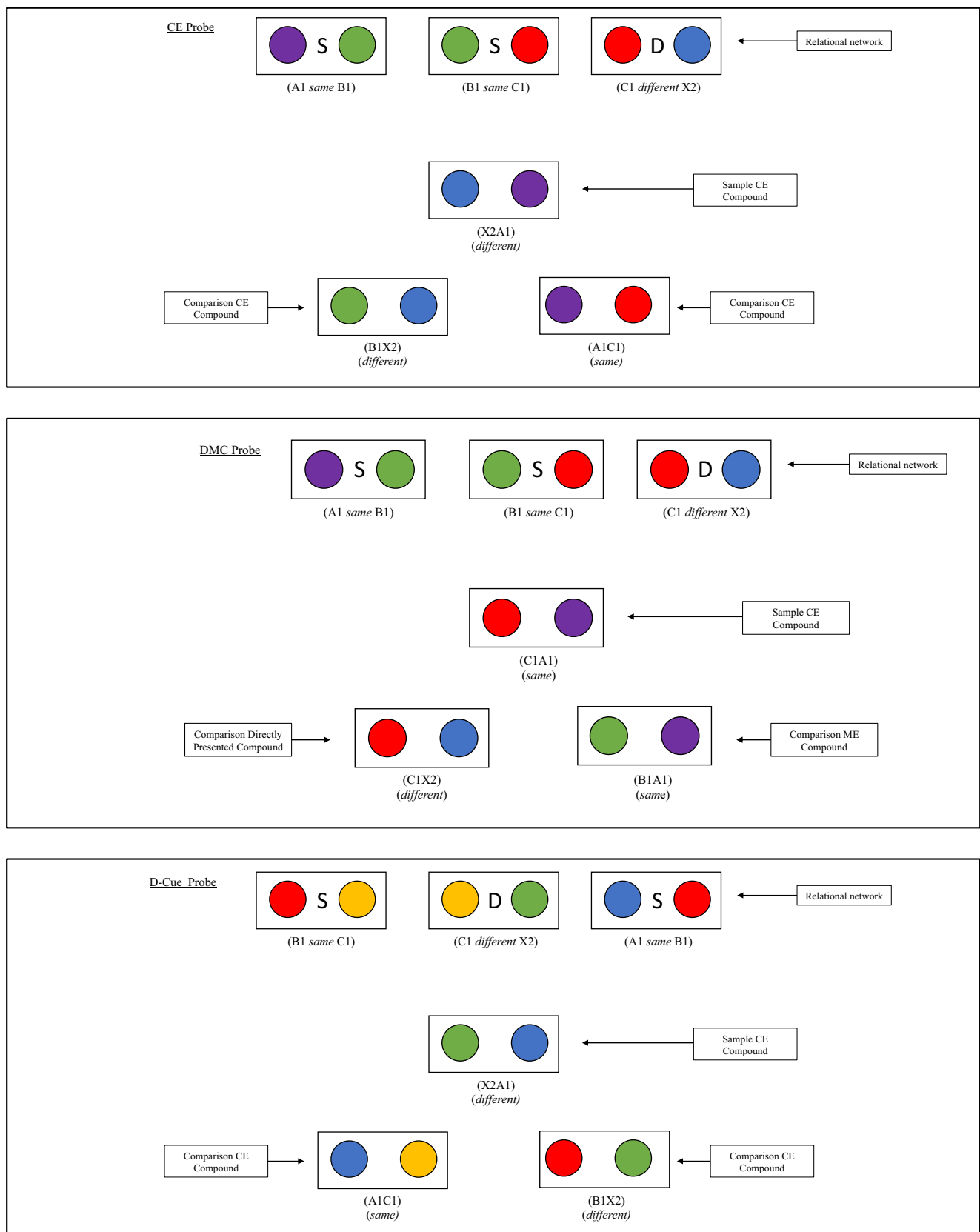


Fig. 3 Three Analogy Probe Types: CE Probes, DMC Probes, and D-Cue Probes. *Note.* First panel: combinatorially entailed relations; second panel: mixed directly presented, mutually entailed, and combinatorially

entailed relations; third panel: the *difference* cue changes location in the relational network plus combinatorially entailed relations

The critical distinction of the D-Cue Probes was that the format of the relational network did not follow the [A [S] B], [B [S] C], [C [D] D] pattern. Instead, the location of the *difference* cue varied on every trial in a probe set. There were always two *sameness* cues and one *difference* cue in the relational networks but across D-Cue probe trials the *difference* cue could be the first, second, or third relational cue. For example, the difference cue is in the first position in the network A is *different* to B, A is the *same* as C, C is the *same* as D ([A] [D] [B], [A] [S] [C], [C] [S] [D]), whereas in the second position in the network, A is the *same* as B, B is *different* to C, A is the *same* as D ([A] [S] [B], [B] [D] [C], [A] [S] [D]). As in the CE Probes, combinatorial entailment was required in order to derive the relation between the two stimuli in all sample and comparison compound elements (see third panel in Fig. 3).

For all probe types (i.e., CE, DMC, and D-Cue), the participant and researcher looked at the laptop screen, the researcher “read” the relational network to/with the participant, and then said, “Look at this one at the top [points to the sample compound], which one of these [points to comparison compounds] is like this one at the top?” In all trials, both comparison compounds included one of the elements from the sample compound so that it was not possible to respond correctly based solely on nonarbitrary similarity. No feedback was provided for correct or incorrect responding. All probe sets included five trials, and passing criteria required responding correctly on all five trials (100%) the first time the probe was presented, or scoring 100% correct twice consecutively. CE Probe Set 1 was presented in baseline sessions, and it was the first probe set after training commenced, followed by novel CE Probes, DMC Probes, novel CE Probes again, and D-Cue Probes, in that order. Failed D-Cue Probes were not followed by more training sessions. Maintenance probes for all three probe types were administered 1 month after each participant’s last probe session.

The training condition itself included two phases (Training Phase 1 and Phase 2). A modified version of the Greer and Ross’s (2008) decision-making protocol was followed during the training condition. If Phase 1 training data showed three increasing data paths, then training in Phase 1 continued. If Phase 1 data showed three stable or three descending data paths, then Phase 2 training would be implemented. No participants required Phase 2 training.

Training sets included the same relational network structure as in the CE Probes (i.e., [A [S] B], [B [S] C], [C [D] D]), but with novel stimuli (i.e., different colored monochromatic circles). Each training set included two PowerPoint slides; the first slide showed the relational network at the top of the screen, and the second slide showed the analogical stimuli below the relational network. Upon presentation of the relational network (the first slide), the researcher and participant read the

relational network together. After reading the network, the second slide with the analogical stimuli was presented. In all trials, comparison compounds included one of the elements from the sample compound such that it was not possible to respond correctly based on nonarbitrary relations. First, the researcher asked the participant to derive the combinatorially entailed (CE) relation for the stimuli in each compound element (the sample and the two comparisons) by pointing to one of the compound elements and asking, “Is A the *same* or *different* to D?” Correct responses included identifying the relation between the stimuli in the compound by saying “same” or “different.” Feedback was provided for each response; for correct responses verbal praise and stickers were immediately delivered, and for incorrect responses the researcher provided the correct response and re-presented the question.

Following the three CE relational questions, the researcher presented the instruction for the analogy task, “Now look at this one at the top (points to the sample), which one of these (points to comparisons) is like the one at the top?” Correct responses included either pointing to the correct comparison compound element or vocally stating the two colors of the correct comparison compound element. Correct responses were consequated with verbal praise and a sticker, and CE Probe Set 1 was re-presented. Incorrect training responses were followed by a correction procedure including: (1) repeating the combinatorially entailed relational questions for all compound elements in the analogy, and (2) re-presenting the analogical question plus pointing to the correct comparison compound element and saying, “this one is like the one at the top.” Following an incorrect trial, another training trial was presented until the participant emitted a correct response, and then the CE Probe was re-presented. If the participant failed the CE Probe, another training set was presented, and the training and probing procedure was repeated until all probe types had been administered.

Interobserver Agreement

Procedural fidelity checks and interobserver agreement (IOA) were determined for baseline, probes, and training conditions by a trained research assistant. Procedural fidelity was assessed through the use of a fidelity checklist in which each trial in each condition was scored as either correct or incorrect; correct presentation required adherence to all relevant procedural criteria based on condition and trial type including presentation and use of the appropriate feedback. Procedural fidelity was assessed for 32% of all trials and was 100%. Interobserver agreement was calculated on a trial-by-trial bases for each probe and training trial. Interobserver agreement was assessed for 43% of all trials and was 100%.

Results

Preassessment

Eight children were given the preassessment measure. Four of these children did not pass the relational section of this test and thus did not proceed to the analogical section of the test. Of the four children who passed the relational section, three of those went on to take part in the multiple baseline training.

In what follows we describe in more detail the preassessment performance of the three children who went on to be trained in the multiple baseline training (see Table 2 for a summary of participant preassessment performance). P1 scored 11/14 (79%) correct on the last set of combinatorial entailment and compound identification trials, 14/15 (93%) correct on the combinatorial entailment relational section of prompt Level 1, and 3/5 (60%) correct on the analogical section of prompt Level 1; 2/5 (40%) on prompt Level 2; and 2/5 (40%) on prompt Level 3. Hence P1 obtained 86% correct on the relational section and 47% correct on the analogical section of the preassessment measure. P2 scored 14/14 (100%) correct on the last set of combinatorial entailment and compound identification trials, 15/15 (100%) correct on the combinatorial entailment relational section of prompt Level 1, and 2/5 (40%) on the analogical section of prompt Level 1; 0/5 (0%) on prompt Level 2; and 1/5 (20%) on prompt Level 3. Hence P2 obtained 100% correct on the relational section and 20% correct on the analogical section of the preassessment measure. P3 scored 14/14 (100%) correct on the last set of combinatorial entailment and compound identification trials, 14/15 (93%) correct on the combinatorial entailment relational section of prompt Level 1, and 2/5 (40%) on the analogical section of prompt Level 1; 2/5 (40%) on prompt Level 2; and 1/5 (20%) on prompt Level 3. Hence P3 obtained 96.5% correct on the relational section and 33% correct on the analogical section of the preassessment measure.

Training

In the multiple baseline training condition, P1 scored 100% correct on her first training session and scored

60% on the following CE Probe Set 1. After three training sessions, P1 scored 100% on CE Probe Set 1 twice consecutively, and 100% on CE Probe Set 2. P1 scored 80% correct on the first generalization probe session, DMC Probe Set 1. P1's responses decreased to 0% correct during the training session after the first DMC probe, and she required five more training sessions before scoring 100% on two consecutive DMC Set 1 probes and one novel DMC probe. On the first maintenance check, P1 scored 100% on CE Probe Set 3. P1 scored 60% on the D-cue probe. In maintenance testing, P1 scored 100% on the CE probe, 100% on the DMC probe, and 60% on the D-cue probe (see Fig. 4).

P2 scored 100% correct on his first training session and 40% on the subsequent CE probe. After three training sessions, P2 scored 100% on CE Probe Set 1 twice consecutively, and 100% on CE Probe Set 2. P2 scored 100% on the first DMC probe and hence did not require another DMC probe. P2 subsequently began to exhibit behavioral issues that interfered with his participation, and as a result he did not complete the D-cue probe after training. However, he completed all three probe types at 1-month maintenance testing and scored 100% on CE Probe Set 3, 100% on the novel DMC probe, and 60% on the D-cue probe.

P3 scored 100% correct on his first training session and scored 60% on the following CE Probe Set 1. After four training sessions, Participant P3 scored 100% on CE Probe Set 1 twice consecutively, and 100% on CE Probe Set 2. Participant P3 scored 80% correct on DMC Probe Set 1 and required three training sessions before scoring 100% twice consecutively on DMC Probe Set 1 and 100% on a novel DMC probe. Following the DMC probes, P3 scored 100% on the maintenance CE Probe Set 3, and 60% on the D-cue probe. During the 1-month follow up maintenance testing, P3 scored 100% correct on all three probes.

Summary

Three out of eight children given preassessment testing met criteria and hence were admitted into multiple baseline

Table 2 Preassessment Relational and Analogical Scores

Participant	Relational Score %	Prompt Level 1		Prompt Level 2	Prompt Level 3	Inclusion Criteria: Mean Scores	
		Relational %	Analogy %	Analogy %	Analogy %	Relational <i>M</i>	Analogy <i>M</i>
P1	79	93	60	40	40	86	47
P2	100	100	40	0	20	100	20
P3	100	93	40	40	20	96.5	33

Note. Inclusion criteria required passing the relational test at 80% or above and failing the analogy test.

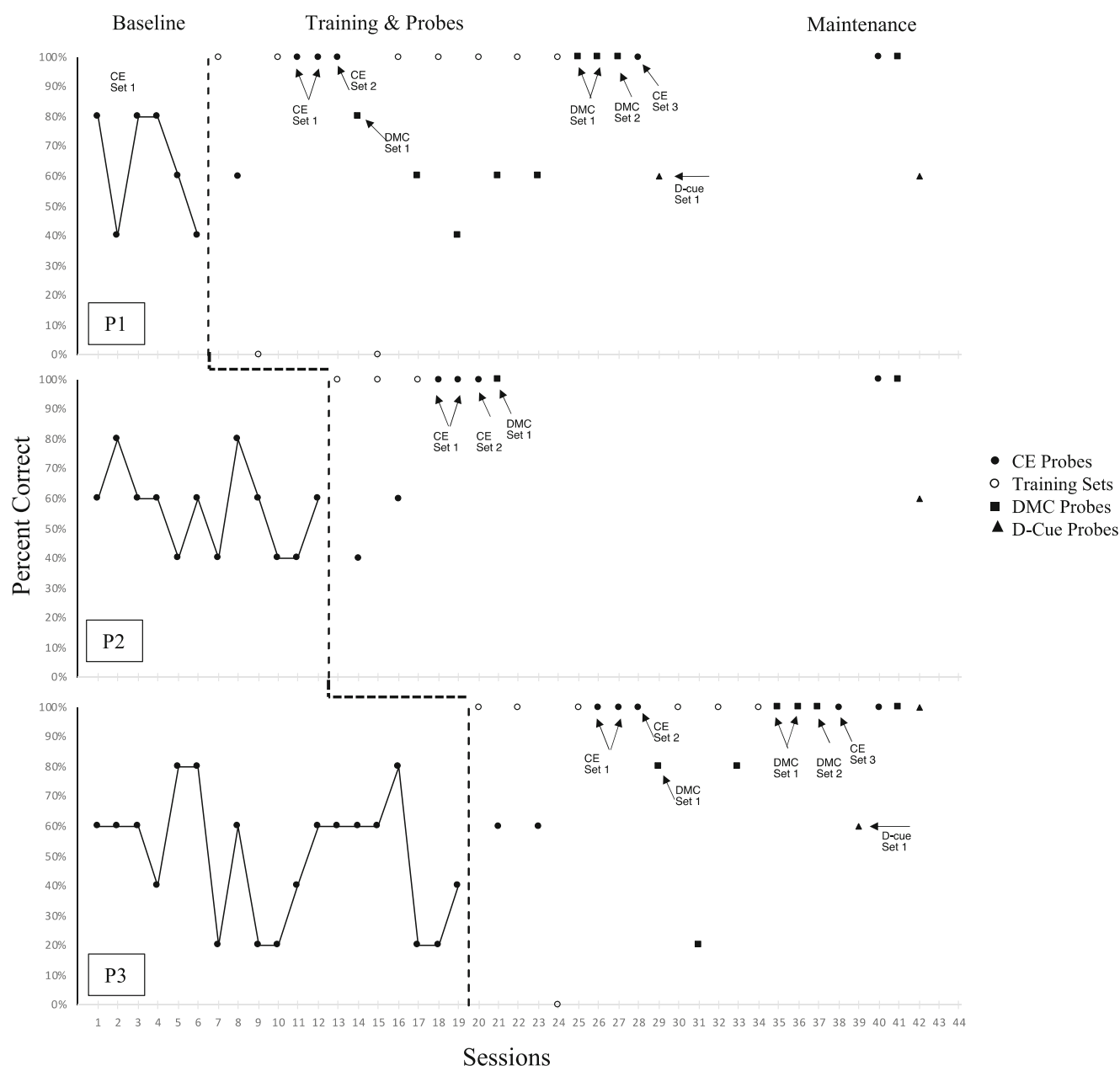


Fig. 4 Participant Responding in Baseline and Training Sessions. *Note.* CE Probes: combinatorially entailed analogical responses; DMC Probes: trained, mutually entailed, and combinatorially entailed analogical

responses; D-cue Probes: the contextual cue for *difference* varied in location within the relational network

training. Following analogy training, all three successfully showed analogical responding during CE Probe sets, including the original CE Probe Set 1 used during baseline testing, as well as a novel CE Probe Set 2. All participants scored 100% correct on the DMC probes. P1 and P3 both scored 60% correct on the D-cue probes, whereas P2 did not complete the D-cue Probe following training. At 1-month maintenance probes, all three participants scored 100% correct on a novel CE Probe set and a novel DMC Probe set, whereas all participants, including P2, scored only 60% correct on the D-Cue Probe.

Discussion

As a key repertoire in complex human behavior, analogy is evident in scientific, mathematical, and problem-oriented domains in addition to being ubiquitous in everyday language (Gentner et al., 2001). Furthermore, reasoning by analogy is central to children's everyday learning, including language and cognitive development (Holyoak & Thagard, 1995; Richland et al., 2006). Despite this, the acquisition of analogy in young children remains understudied within the applied behavioral sciences. The RFT account of analogy as derived

relating of relations allows for a functional analysis of analogical responding, which facilitates training this repertoire. The present study aimed to extend previous RFT-based research in analogy in young children including several studies by Carpentier et al. who provided the first RFT-based assessment of analogy as the derived relating of relations as well as Kirsten and Stewart (2021) who assessed arbitrary relational responding including analogy in young children using a newly developed relational assessment format.

After direct training in relating combinatorially entailed relations, all three participants in the present study showed analogical responding according to RFT's conception of analogy as the derived relating of relations. This extends previous work by Carpentier et al. (2002, 2003) who were the first to use this RFT approach to examine analogy in young children. In Carpentier et al.'s studies, 5-year-old children initially failed to derive equivalence–equivalence relations without additional prompting. In particular, the 5-year-old children required pretesting with compound–compound matching tasks involving trained relations (i.e., not derived relations) before they successfully passed the derived compound relations (BC–BC) test. These results indicated that 5-year-old children are capable of analogical reasoning but in this case, they required prompting before doing so. The present data support Carpentier et al.'s findings that 5-year-old children are capable of analogical responding. However, the present study extends Carpentier et al. Rather than simply facilitating the emergence of analogy through prompting procedures, the present data extend this previous research by directly training analogy through multiple exemplar training using a controlled multiple baseline design such that the participants of the study could thereafter demonstrate analogical responding without additional prompting procedures being needed.

The results from the multiple baseline showed that the CE analogy training procedure was an effective intervention for training analogy and eliciting generative responding as shown by the generalization data. All participants passed baseline CE Probe Set 1 as well as novel CE probe sets following CE analogy training. Furthermore, correct analogical responding generalized to the DMC probes. Previous RFT work on relating relations (e.g., Barnes et al., 1997; Carpentier et al., 2002) typically involved testing and training at least two sets of *combinatorially* entailed relations. The present work included directly presented, mutually entailed, and combinatorially entailed relations to provide evidence that participants were generalizing analogical responding across different types of relational derivation rather than limiting their responses to the CE relation only.

None of the participants scored 100% correct on the first presentation of the D-cue generalization probes. However, during 1-month follow-up maintenance testing, one participant scored 100% correct on all probe types, including the D-Cue probe. In the D-Cue Probes, the format of the relational

network did not follow the [A [S] B], [B [S] C], [C [D] D] pattern. There were always two *sameness* cues and one *difference* cue in the relational networks but across D-Cue probe trials the *difference* cue could be the first, second, or third relational cue. As in the CE Probes, combinatorial entailment was required in order to derive the relation between the two stimuli in all sample and comparison compound elements. It is unclear why participants did not score higher on the D-Cue probes. A limitation of the present study is that additional training was not provided following the failed probe. The D-Cue probe was used only to examine whether generalization to such a probe might be seen; it remains unknown if participants might have passed the D-Cue probes with more CE analogy training. Future researchers could investigate whether generalization would eventually extend to the D-Cue probe (as well as other novel probes) with more CE analogy training. Kirsten and Stewart (2021) indicated that 5-year-old participants may not have had sufficient practice with arbitrary relational responding to relate relations successfully. Thus, more practice with multiple exemplars in relational responding may be necessary for generalization to occur in more complex relational responding such as in relating relations. Using the REP format, future researchers could examine the effects of additional arbitrary relational training on analogical responding.

Previous RFT studies have successfully shown that many different patterns of relational framing can be tested and trained via multiple exemplar training (e.g., Belisle et al., 2016; Berens & Hayes, 2007; Cassidy et al., 2011, 2016; Mulhern et al., 2018). These include not only simple frames such as comparison and opposition (e.g., Berens & Hayes, 2007; Cassidy et al., 2011, 2016) but also more complex patterns of framing such as hierarchy (Mulhern et al., 2018) and deixis (Belisle et al., 2016). The results from the present study provide evidence that derived relating of relations can also be trained using multiple exemplar training just like the others listed and as such this is yet further evidence in support of the RFT thesis.

The present data also constitute an important addition to the literature on analogical relations beyond behavior analysis. Cognitive researchers have studied the acquisition of analogical responding in young children since Piaget's early work in the 1970s (e.g., Piaget et al., 1977, 2001; Goswami, 1989, 1996, 2001; Rattermann & Gentner, 1998; Richland et al., 2006). However, as has been argued in previous RFT literature, this approach to analogical development has failed to lead to a functional analytic model of this behavior that might lend itself to training this repertoire. Moreover, within the cognitive literature, the age at which analogical responding develops in young children remains a debate among researchers (Richland et al., 2006), and flaws in testing methodologies have hindered progress in testing and training analogy in this population. For example, Goswami and Brown (1989)

provided data of analogical responding in 3-year-old children. However, Rattermann and Gentner (1998) provided similarity ratings for the stimuli used to test for analogy in Goswami and Brown (1989) and concluded that children younger than 5 responded correctly based on object similarity rather than on analogical relations. Another vein of cognitive research has examined the use of analogy to facilitate learning in other domains, but training analogical responding itself was not included (e.g., Brown & Kane, 1988; Polya, 1945/2004; Richland & McDonough, 2010). Thus, within the cognitive literature, the development of analogical responding in young children has been theorized and examined, and the use of analogies to solve daily problems encountered in life or at school has been examined, but there is a dearth of experimental work on the effective training of analogy in young children. In contrast, RFT's functional analytic definition of analogy has allowed a reliable methodology for targeting analogical reasoning in this population.

One important feature of the present study is that analogical responding was tested and trained using a novel RFT-based protocol, and in particular, an adaptation of the RFT-based protocol used in Kirsten and Stewart (2021). The utilization of shapes across all trials obviated the need to control for participants' previous experience and knowledge as well as controlled for stimulus consistency across trials. The results of this study suggest the applied and experimental potential of this format. For example, by using monochromatic circles and single-lettered contextual cues, the REP format makes it easy to produce multiple training and testing sets suitable for young children and prereaders, which in turn makes it possible to directly train children on complex relations with many exemplars. Moreover, this format does not require any relational pretraining but instead allows for faster testing (with multiple versions of tests) than is possible with the tricky methodological issues in MTS procedures. And finally, a participant's previous experience and knowledge is not an experimental confound. These favorable variables of the REP format permitted a multiply controlled study in which we could target analogy testing and training directly while maintaining experimental control. That is, the REP format afforded us with quick and effective stimulus control allowing us to implement a multiple baseline design to examine the efficacy of multiple exemplar training to establish the core repertoire. For example, once participants were trained on the format of the REP in the preassessment, we were able to immediately implement analogical baseline testing across all participants, and then implement controlled training and testing conditions with novel sets of stimuli. In addition, the REP format provided design flexibility for different generalization probe types. In contrast, the time-consuming and laborious pretraining required in match-to-sample procedures for example would have made multiple testing and training conditions in a controlled multiple baseline design unfeasible.

One innovative aspect of the present study is that it involved using an REP procedure involving relational "statements" with children at an age at which reading is as yet only beginning to be acquired as a repertoire. As we stated in the procedure, reading involved vocally identifying the stimuli and relational cues in the relational network in sequence from left to right, similar to textual reading. In the preassessment and in the multiple baseline conditions, the assessor first read the relational network to/with the participant and then asked the trial questions. If the participant requested or initiated independent reading when presented with the relational network, the researcher did not read with the participant but simply allowed them to read alone. The latter aspect of independent reading itself is one possible variable that could be examined further in future research. For example, it might be interesting to examine the effects on analogical (or other forms of) relational responding after training participants to read the relational network independently if they are unable to do so as opposed to reading the network for or alongside them as was the default in this study.

Future researchers could also increase the number of stimuli in the relational network to include more than four relata (i.e., the stimuli being related). A larger array would more closely replicate the analogical work by Carpentier et al. (2002) in which participants were trained to identify nine arbitrary pictures as being equivalent or nonequivalent and were then tested for equivalence–equivalence and nonequivalence–nonequivalence (i.e., relating relations). Future researchers could introduce multiple relational networks by increasing the array of monochromatic circles and using only the *sameness* cue, S, to relate particular circles to one another. For example, as in Carpentier et al. an array of nine circles could be divided into three 3-member sets; participants would refer to the relational networks to determine equivalence or nonequivalence between compound elements, followed by equivalence–equivalence or nonequivalence–nonequivalence testing and training. A larger array would allow for more combinatorially entailed sample and comparison compound combinations, thus circumventing a potential limitation in the present study. That is, the relational networks were composed of two *sameness* cues and one *difference* cue (e.g., Red S Blue, Blue S Green, Green D Yellow) and therefore permitted only combinatorially derived *difference* relations. Thus, the compound elements in the CE and D-cue probe trials had to be *difference* relations. *Sameness* relations were possible via directly presented, mutually entailed, or combinatorially entailed relations as in the DMC probes. It is possible that participants learned to pick the *difference* comparison compound instead of relating *difference-difference* relations. However, during the DMC probes it was possible for both *sameness* and *difference* relations to be tested because the stimuli in the sample and comparison compound elements included directly trained, mutually entailed, and combinatorially entailed relations. All participants passed the DMC probes, which

suggests that they were responding to analogical relations rather than just selecting the *difference* compound. However, future work could further examine testing and training analogical relations as relating relations by increasing the number of relational networks.

The REP format could also be extended to test and train analogical relations beyond coordination and distinction. Lipkens and Hayes (2009) successfully looked at multiple relations (i.e., sameness, difference, comparison, and opposition) in analogy in adult participants. Lipkens and Hayes tested across selection-based and topography-based tasks including, selecting the correct relata, selecting the correct relational cue, and producing the correct relata. However, the Lipkens and Hayes procedure required extensive pretraining as well as reading and writing skills. For young children, it might be possible to test and train multiple relations and topography-based responses using variations of the REP format used in the present study. For example, in Lipkens and Hayes's *selecting relata task* the participant was given the sample, a relational cue, and two comparisons; participants had to select the correct comparison based on the relational cue; in the *selecting the correct relation task*, the participant had to select the correct relational cue given the sample and the comparison stimulus; and in the *producing relata task*, the participant had to produce the correct relata given the sample stimulus and relational cue. The REP format would support these various tasks and make them accessible to young children. For example, in producing the relata tasks, the response could include shading in a blank circle with the correct color, and in selecting the relation tasks, the same label format (i.e., *S* or *D*) used in the current study could be applied and extended to other relations (see, for example Kirsten & Stewart, 2021).

Future researchers could also examine the effects of training sameness relations on the emergence of other relations. For example, once participants have been trained in coordinate analogical responding, future researchers could probe other types of relations (e.g., comparative-comparative) for generalization. Kirsten and Stewart (2021) found that coordinate analogical responding was acquired before comparative, opposite, temporal, and hierarchical analogical responding, thus future research could also use the present format to test and train children to relate these other relations.

Considering the relevance of analogy to intelligence, future researchers could investigate the generalized effects of training analogical responding on socially valid measures. For example, future dependent variables could include mainstream analogy tests, academic achievement tests, or standardized tests of cognitive performance. Furthermore, because relating relations allows response classes to affect other response classes, future researchers could also investigate whether training analogical responding in young children facilitates the emergence of developing repertoires such as reading or mathematics. For example, progress in reading requires understanding increasingly complex

and subtle analogical/metaphorical relationships whereas mathematics learning can also benefit from a repertoire of analogical responding (e.g., Richland et al., 2004).

The present study adds to the limited behavioral research on analogical responding in young children, and contributes further evidence that 5-year-old children can be trained in analogical responding. Moreover, by offering a precise, functional-analytic model of analogy, the present study has arguably shed additional light into this phenomenon beyond that provided by mainstream cognitive psychological work. This work provides further evidence that analogy tends to be acquired around the age of 5 but goes beyond previous work in showing how multiple exemplar training of the core pattern involved might accelerate this acquisition. Considering the ubiquity of analogical responding in everyday life, more research regarding its development and training in young children both with and without language delays is merited. Finally, the potential of the REP format used in this study to test and train young children in complex relational responding, such as analogy, is promising and lends itself to further investigation of its research and applied utility.

Code Availability Not applicable

Data Availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

Ethical Approval The study was approved by the NUI Galway Research Ethics Committee. All procedures performed in the studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent to Participate Informed consent was obtained from all individual participants included in the study.

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